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Integral points for groups of multiplicative type

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To Professor Kezheng Li on his 65-th birthday

Abstract

We construct a finite subgroup of Brauer–Manin obstruction for detecting the existence of integral points on integral models of homogeneous spaces of linear algebraic groups of multiplicative type. As an application, the strong approximation theorem for linear algebraic groups of multiplicative type is established. Moreover, the sum of two integral squares over some quadratic fields is discussed.
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0. Introduction

The integral points on homogeneous spaces of semi-simple and simply connected linear algebraic groups of non-compact type were studied by Borovoi and Rudnick in [1] and homogeneous spaces of connected semi-simple linear algebraic groups of non-compact type by Colliot-Thélène and the second author in [2] using the strong approximation theorem and the Brauer–Manin obstruction. Recently, Harari [5] showed that the Brauer–Manin obstruction

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accounts for the nonexistence of integral points. Colliot-Thélène noticed that a finite subgroup of the Brauer group is enough to account for the nonexistence of integral points by compactness arguments. These results are nonconstructive: they do not say which finite subgroup to use. In our previous paper [16], we construct such finite groups for multi-norm tori explicitly. In this paper, we extend this construction for linear algebraic groups of multiplicative type.

The paper is organized as follows. In Section 1, we give an explicit construction of the so called admissible groups (see Definition 1.9) and establish a criterion for the existence of the integral points in terms of admissible groups. In Section 2, we translate such admissible groups into the finite Brauer–Manin obstruction. As an application, we establish the strong approximation theorem for a group of multiplicative type in Section 3. In Section 4, we discuss the sum of two integral squares over some quadratic fields.

Notation and terminology are standard if not explained. Let F be a number field, o_F be the ring of integers of F, Ω_F be the set of all primes in F and ∞_F be the set of all infinite primes in F. For simplicity, we write $p < \infty_F$ for $p \in \Omega_F \setminus \infty_F$. For any finite set $S_0 \subset \Omega_F \setminus \infty_F$, the ring of S_0-integers of F is defined as

$$o_{S_0} = \{x \in F : |x|_p \leq 1 \text{ for all } p < \infty_F \text{ and } p \notin S_0\}.$$

Let F_p be the completion of F at p and o_{F_p} be the local completion of o_F at p for each $p \in \Omega_F$. Write $o_{F_p} = F_p$ for $p \in \infty_F$. We also denote the adele ring (resp. the idele group) of F by $\mathbb{A}_F$ (resp. $\mathbb{I}_F$) and

$$F_\infty = \prod_{p \in \infty_F} F_p.$$

Let $\bar{F}$ be a fixed algebraic closure of F. A group G of multiplicative type over F is defined as a closed subgroup of a torus over F. Let $\hat{G} = Hom_{\bar{F}}(G, \mathbb{G}_m)$ be the character of G. Then the functor $G \mapsto \hat{G}$ is an anti-equivalence between the category of F-groups of multiplicative type and the category of finitely generated abelian groups with the continuous action of $Gal(\bar{F}/F)$ (see [13]). Moreover, G is a torus if and only if $\hat{G}$ is free. For any positive integer k, the finite group scheme $G[k]$ over F stands for the kernel of the multiplication by k over G.

Let $\mathbf{X}$ be a separated o_F-scheme of finite type whose generic fiber X_F is a trivial torsor of G. The obvious necessary condition for $\mathbf{X}(o_F) \neq \emptyset$ is

$$\prod_{p \in \Omega_F} \mathbf{X}(o_{F_p}) \neq \emptyset \tag{0.1}$$

which is assumed throughout this paper. The Brauer group $Br(X_F)$ of X_F is defined as

$$Br(X_F) = H^2_{et}(X_F, \mathbb{G}_m) \quad \text{and} \quad Br_1(X_F) = ker[Br(X_F) \to Br(\bar{X})]$$

where $\bar{X} = X_F \times_F \bar{F}$. Since the image of $Br(F)$ induced by the structure morphism lies in $Br_1(X_F)$, one defines

$$Br_a(X_F) = coker[Br(F) \to Br_1(X)].$$

For any subset $\mathfrak{s}$ of $Br_a(X_F)$, one can define the integral Brauer–Manin set with respect to $\mathfrak{s}$ as (see [2])

$$\left(\prod_{p \in \Omega_F} \mathbf{X}(o_{F_p})\right)^{\mathfrak{s}} = \left\{(x_p) \in \prod_{p \in \Omega_F} \mathbf{X}(o_{F_p}) : \sum_{p \in \Omega_F} inv_p(s(x_p)) = 0, \ \forall s \in \mathfrak{s}\right\}.$$

1. Construction of X-admissible groups

In this section, we extend the construction of **X**-admissible groups in [16] for multi-norm tori to groups of multiplicative type. By using the anti-equivalence between the category of F-tori and the category of free $\mathbb{Z}$-modules of finite rank with the continuous action of $Gal(\bar{F}/F)$, one has the following lemma which should be standard.

Lemma 1.1. *Let T_1 and T_2 be two tori over F and $\phi : T_1 \to T_2$ be a surjective morphism of tori. Then there is a morphism of tori*

$$\psi : T_2 \to T_1 \quad \text{such that } \phi \circ \psi = [l]$$

for some positive integer l.

Proof. Since the category of F-tori is anti-equivalent to the category of free $\mathbb{Z}$-modules of finite rank with the continuous action of $Gal(\bar{F}/F)$, one has the injective $Gal(\bar{F}/F)$-module homomorphism $\hat{\phi} : \hat{T}_2 \to \hat{T}_1$. Extending this map over $\mathbb{Q}$, one obtains the injective $Gal(\bar{F}/F)$-module homomorphism

$$\hat{\phi}_{\mathbb{Q}} : \hat{T}_2 \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow \hat{T}_1 \otimes_{\mathbb{Z}} \mathbb{Q}.$$

Since a representation of a compact group over a field of characteristic 0 is semi-simple (Maschke's Theorem), there is a $Gal(\bar{F}/F)$-module homomorphism

$$\hat{\psi'}_{\mathbb{Q}} : \hat{T}_1 \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow \hat{T}_2 \otimes_{\mathbb{Z}} \mathbb{Q}$$

such that $\hat{\psi'}_{\mathbb{Q}} \circ \hat{\phi}_{\mathbb{Q}} = 1$. Since $\hat{T}_1$ is of finite rank, there is a positive integer l such that $l\hat{\psi'}_{\mathbb{Q}}(\hat{T}_1) \subseteq \hat{T}_2$. Then the homomorphism from T_2 to T_1 associated to $l\hat{\psi'}_{\mathbb{Q}}$ is as required. □

Fix a finite Galois extension E/F such that $Gal(\bar{F}/E)$ acts on $\hat{G}$ trivially. Let M be a free $\mathbb{Z}$-module of finite rank such that $f : M \to \hat{G}$ is a surjective homomorphism. One can extend this map to the $Gal(\bar{F}/F)$-module surjective homomorphism

$$\mathbb{Z}[Gal(E/F)] \otimes_{\mathbb{Z}} M \to \hat{G}, \quad \sigma \otimes m \mapsto \sigma(f(m))$$

where $Gal(\bar{F}/F)$ acts on $\mathbb{Z}[Gal(E/F)] \otimes_{\mathbb{Z}} M$ as follows

$$\tau \circ (\sigma \otimes m) = \bar{\tau}\sigma \otimes m$$

for $\tau \in Gal(\bar{F}/F)$ and $\sigma \otimes m \in \mathbb{Z}[Gal(E/F)] \otimes_{\mathbb{Z}} M$. By the anti-equivalence between the category of multiplicative groups over F and the category of finitely generated $\mathbb{Z}$-modules with the continuous action of $Gal(\bar{F}/F)$, there is an F-torus T such that

$$0 \to G \xrightarrow{\lambda} \prod_1^n R_{E/F}(\mathbb{G}_m) \xrightarrow{\nu} T \to 0 \tag{1.2}$$

where $n = rank_{\mathbb{Z}}(M)$.

Applying Lemma 1.1 to (1.2), one can fix a homomorphism

$$\mu : T \to \prod_1^n R_{E/F}(\mathbb{G}_m)$$

such that $\nu \circ \mu = [l]$ for some positive integer l.

Since all morphisms are of finite type, we fix a finite subset $S_0 \subset \Omega_F \setminus \infty_F$ such that

(1) the groups G, $\prod_1^n R_{E/F}(\mathbb{G}_m)$ and T over F extend to the group schemes $\mathbf{G}$, $\mathbf{R}_{E/F}(\mathbb{G}_m)$ and $\mathbf{T}$ over o_{S_0} such that (1.2) extends to

$$0 \longrightarrow \mathbf{G} \longrightarrow \prod_1^n \mathbf{R}_{E/F}(\mathbb{G}_m) \longrightarrow \mathbf{T} \longrightarrow 0$$

over o_{S_0} and

$$\mathbf{R}_{E/F}(\mathbb{G}_m)(o_{F_p}) = o_{E_p}^{\times}$$

for $p \notin S_0$.

(2) the trivial torsor X_F of G over F extends to the trivial torsor $\mathbf{X}_{S_0} = \mathbf{X} \times_{o_F} o_{S_0}$ of $\mathbf{G}$ over o_{S_0}.

(3) the homomorphism μ also extends to $\mathbf{T} \to \prod_1^n \mathbf{R}_{E/F}(\mathbb{G}_m)$ over o_{S_0}.

(4) any l-torsion point of $T(F_p)$ is contained in $\mathbf{T}(o_{F_p})$ for all $p \notin S_0$.

For the above torus T, the class group of T with respect to S_0 can be defined as follows

$$cls(T, S_0) = T(\mathbb{A}_F) \Big/ \left[T(F) + \left(\prod_{p \in S_0} T(F_p) \times \prod_{p \notin S_0} \mathbf{T}(o_{F_p}) \right) \right]$$

and this group $cls(T, S_0)$ is finite by Theorem 5.1 in [12]. Denote $h(T, S_0) = \sharp cls(T, S_0)$.

Let A be a finite group scheme over F. One defines

$$Ш_F^1(A) = ker \left[H^1(F, A) \longrightarrow \prod_{p < \infty_F} H^1(F_p, A) \right].$$

Lemma 1.3. *For any torus T over F, there is a positive integer $c = c(F, T)$ such that $Ш_F^1(T[k])$ is killed by c for all positive integers k.*

Proof. Let E be a finite Galois extension over F such that

$$T_E := T \times_F E \cong \prod_1^n \mathbb{G}_{m,E}$$

for some positive integer n. Therefore $T_E[k] = \prod_1^n \mu_k$. Using the inflation–restriction exact sequence of group cohomology, together with its functoriality properties, one obtains the following commutative diagram

$$\begin{array}{ccccc} H^1(E/F, T(E)[k]) & \longrightarrow & H^1(F, T[k]) & \longrightarrow & H^1(E, T_E[k]) \\ & & \downarrow & & \downarrow \\ & & \prod_{p < \infty_F} H^1(F_p, T[k]) & \longrightarrow & \prod_{P < \infty_E} H^1(E_P, T_E[k]). \end{array}$$

By (9.1.3) Theorem in [9], one knows that $Ш_E^1(T_E[k])$ is killed by 2. If $\mu(E)$ is the order of the set of all roots of unity contained in E, then $H^1(E/F, T(E)[k])$ is killed by the greatest common divisor of $\mu(E)$ and $[E : F]$. One can take $c = 2(\mu(E), [E : F])$. □

From the above proof, one can further obtain a uniform bound for $Ш_F^1(T[k])$ for all positive integers k. The above result is good enough for our application.

Lemma 1.4. *If $T \cong \prod_1^d \mathbb{G}_m$ for some positive integer d, then*

$$T(F) \cap \left[\prod_{p \in \infty_F} T(F_p) \times \prod_{p \notin \infty_F} [2k]T(F_p) \right] \subseteq [k]T(F)$$

for any positive integer k.

Proof. Any element

$$x \in T(F) \cap \left[\prod_{p \in \infty_F} T(F_p) \times \prod_{p \notin \infty_F} [2k]T(F_p) \right]$$

can be written as $x = (x_1, \ldots, x_d)$ where $x_i \in F^\times$ and $x_i \in (F_p^\times)^{2k}$ for all $p < \infty_F$ with $1 \leq i \leq d$. Applying (9.1.3) Theorem in [9] for μ_{2k}, one obtains that $x_i^2 \in (F^\times)^{2k}$ for $1 \leq i \leq d$. There is $y_i \in F^\times$ such that $x_i = y_i^k$ or $x_i = -y_i^k$ with $1 \leq i \leq d$.

Suppose $x_i = -y_i^k$ for some $1 \leq i \leq d$. Let $k = 2^s k_1$ with $2 \nmid k_1$ and $\zeta_{2^{s+1}}$ be a primitive 2^{s+1}-th root of unity. Then $\zeta_{2^{s+1}} \in F_p^\times$ for all $p < \infty_F$. This implies that all primes $p < \infty_F$ are split completely in the abelian extension $F(\zeta_{2^{s+1}})/F$. By the Chebotarev density theorem, one concludes that $\zeta_{2^{s+1}} \in F$. Therefore $-1 = (\zeta_{2^{s+1}})^k \in (F^\times)^k$ and the proof is complete. □

For a general torus T, one can fix a splitting field K/F of T such that

$$T_K = T \times_F K \cong \prod_1^d \mathbb{G}_{m,K}$$

for some positive integer d. Since K/F is a finite extension, the group of K-rational torsion points of T_K is finite. Fix a positive integer t such that t kills all K-rational torsion points of T_K.

Lemma 1.5. *Let T be a torus over F as above and k be a positive integer. Then there is a finite subset S_1 of $\Omega_F \setminus \infty_F$ containing S_0 such that*

$$T(F) \cap \left[\prod_{p \in S} [c \cdot 2t \cdot h(T, S_0) \cdot k]T(F_p) \times \prod_{p \notin S} \mathbf{T}(o_{F_p}) \right] \subseteq [k]T(F)$$

for any finite subset $S \supseteq S_1$, where $c = c(F, T)$ is as in Lemma 1.3 and t is as above.

Proof. Let U_0 be a finite subset of $\Omega_K \setminus \infty_K$ such that the ring o_{U_0} of U_0-integers of K is the integral closure of o_{S_0} inside K. Let U be a finite subset of $\Omega_K \setminus \infty_K$ containing U_0 such that

$$\mathbf{T}_{U_0}(o_U) = \prod_1^n o_U^\times$$

where $\mathbf{T}_{U_0} = \mathbf{T} \times_{o_{S_0}} o_{U_0}$. By the Dirichlet unit theorem, one obtains that $\mathbf{T}_{U_0}(o_U)$ is a finitely generated abelian group. Since $\mathbf{T}(o_{U_0}) = \mathbf{T}_{U_0}(o_{U_0})$ is a subgroup of $\mathbf{T}_{U_0}(o_U)$ and $\mathbf{T}(o_{S_0})$ is a subgroup of $\mathbf{T}(o_{U_0})$, one concludes that $\mathbf{T}(o_{S_0})$ is a finitely generated abelian group. Therefore $\mathbf{T}(o_{S_0})/[c \cdot 2t \cdot k]\mathbf{T}(o_{S_0})$ is finite.

If the coset $\alpha + [c \cdot 2t \cdot k]\mathbf{T}(o_{S_0})$ of $\mathbf{T}(o_{S_0})/[c \cdot 2t \cdot k]\mathbf{T}(o_{S_0})$ satisfies that

$$\alpha \notin [c \cdot 2t \cdot k]T(F_p)$$

for some prime $p < \infty_F$, we fix such a prime p_α. Let $\mathfrak{S}$ be the set consisting of all such p_α and $S_1 = S_0 \cup \mathfrak{S}$.

For any finite set $S \supseteq S_1$ and

$$\alpha \in T(F) \cap \left[\prod_{p\in S} [c \cdot 2t \cdot h(T, S_0) \cdot k] T(F_p) \times \prod_{p\notin S} \mathbf{T}(o_{F_p}) \right],$$

there is $\beta_p \in T(F_p)$ such that $\alpha = [c \cdot 2t \cdot h(T, S_0) \cdot k]\beta_p$ for each $p \in S$. Since β_p can be viewed as an element in $T(\mathbb{A}_F)$ whose p component is β_p and the others are 1, there is $\varpi_p \in T(F)$ such that

$$\varpi_p + [h(T, S_0)]\beta_p \in \left[\prod_{p\in S_0} T(F_p) \times \prod_{p\notin S_0} \mathbf{T}(o_{F_p}) \right]$$

for each finite $p \in S \setminus S_0$. This implies that

$$\gamma = \alpha + [c \cdot 2t \cdot k] \left(\sum_{p\in S\setminus(S_0\cup\infty_F)} \varpi_p \right) \in \mathbf{T}(o_{S_0}).$$

Therefore the coset of $\gamma + [c \cdot 2t \cdot k]\mathbf{T}(o_{S_0})$ of $\mathbf{T}(o_{S_0})/[c \cdot 2t \cdot k]\mathbf{T}(o_{S_0})$ is not one of the above mentioned cosets. One concludes that

$$\alpha \in T(F) \cap \left[\prod_{p\in\infty_F} T(F_p) \times \prod_{p\notin\infty_F} [c \cdot 2t \cdot k] T(F_p) \right].$$

Hence

$$\begin{aligned} &T(F) \cap \left[\prod_{p\in S} [c \cdot 2t \cdot h(T, S_0) \cdot k] T(F_p) \times \prod_{p\notin S} \mathbf{T}(o_{F_p}) \right] \\ &\quad \subseteq T(F) \cap \left[\prod_{p\in\infty_F} T(F_p) \times \prod_{p\notin\infty_F} [c \cdot 2t \cdot k] T(F_p) \right]. \end{aligned}$$

For any element

$$x \in T(F) \cap \left[\prod_{p\in\infty_F} T(F_p) \times \prod_{p\notin\infty_F} [c \cdot 2t \cdot k] T(F_p) \right],$$

there is $y \in T(F)$ such that $[c]x = [c \cdot 2t \cdot k]y$ by Lemma 1.3. Let $z = x - [2t \cdot k]y$. Then $z \in T(F)[c]$. By applying Lemma 1.4 to $T_K = T \times_F K$ over K, one obtains that there is $w \in T(K)$ such that $z = [t \cdot k]w$. Since z is a torsion point of T, one has that w is also a torsion point of T_K. Since t kills all K-torsion points of T_K, one concludes that $z = 0$. The proof is complete. □

Lemma 1.6. *If C_p is an open subgroup of $G(F_p)$ and k is a positive integer, then*

$$\lambda(C_p) \cdot \mu([k]T(F_p))$$

is an open subgroup of

$$\prod_1^n R_{E/F}(\mathbb{G}_m)(F_p) = \prod_1^n E_p^\times$$

for the p-adic topology with $p < \infty_F$.

Proof. Since C_p is an open subgroup of $G(F_p)$ and $\lambda[G(F_p)]$ is a closed subgroup of

$$\prod_1^n R_{E/F}(\mathbb{G}_m)(F_p) = \prod_1^n E_p^\times,$$

there is a positive integer a such that

$$\left[\prod_1^n (1 + p^a o_{E_p})\right] \cap \lambda[G(F_p)] \subseteq \lambda(C_p).$$

Since $\mu \circ \nu$ is a continuous endomorphism of

$$\prod_1^n R_{E/F}(\mathbb{G}_m)(F_p) = \prod_1^n E_p^\times,$$

there is a positive integer $a_1 > a$ such that

$$\mu \circ \nu \left[\prod_1^n (1 + p^{a_1} o_{E_p})\right] \subseteq \prod_1^n (1 + p^a o_{E_p}).$$

By Hensel's lemma, there is a positive integer $a_2 > a_1$ such that

$$(1 + p^{a_2} o_{E_p}) \subseteq (1 + p^{a_1} o_{E_p})^{lk}.$$

For any $x \in \prod_1^n (1 + p^{a_2} o_{E_p})$, there is

$$y \in \prod_1^n (1 + p^{a_1} o_{E_p}) \quad \text{such that } x = y^{lk}.$$

Then

$$x = y^{lk} = [y^{lk}(\mu \circ \nu(y^k))^{-1}](\mu \circ \nu(y^k)).$$

Since

$$\nu[y^{lk}(\mu \circ \nu(y^k))^{-1}] = [l]\nu(y^k) - \nu \circ \mu \circ \nu(y^k) = 0,$$

one concludes

$$y^{lk}(\mu \circ \nu(y^k))^{-1} \in \lambda(C_p).$$

Therefore

$$\prod_1^n (1 + p^{a_2} o_{E_p}) \subseteq \lambda(C_p) \cdot \mu([k]T(F_p))$$

and the proof is complete. □

Since $\mathbf{X}$ is separated over o_F, one can view $\mathbf{X}(o_{F_p})$ as an open subset of $X_F(F_p)$ by the natural map for any $p \in \Omega_F$.

Definition 1.7. Define

$$Stab(\mathbf{X}(o_{F_p})) = \{g \in G(F_p) : \ g\mathbf{X}(o_{F_p}) = \mathbf{X}(o_{F_p})\}$$

for $p \in \Omega_F$.

It is clear that $Stab(\mathbf{X}(o_{F_p}))$ is an open subgroup of $G(F_p)$. Since X_F is a trivial torsor of G over F, one has that

$$Stab(\mathbf{X}(o_{F_p})) = \mathbf{G}(o_{F_p})$$

for almost all p in Ω_F.

Definition 1.8. Define

$$Stab_{\mathbb{A}}(\mathbf{X}) = \prod_{p\in\Omega_F} Stab(\mathbf{X}(o_{F_p})).$$

It is clear that $Stab_{\mathbb{A}}(\mathbf{X})$ is an open subgroup of $G(\mathbb{A}_F)$. The map given by (1.2)

$$\lambda : G \longrightarrow \prod_1^n R_{E/F}(\mathbb{G}_m)$$

induces the homomorphism

$$\lambda : \ G(\mathbb{A}_F) \longrightarrow \prod_1^n \mathbb{I}_E.$$

Definition 1.9. An open subgroup Ξ of $\prod_1^n \mathbb{I}_E$ is called $\mathbf{X}$-admissible if

$$\lambda[Stab_{\mathbb{A}}(\mathbf{X})] \subseteq \Xi$$

and the induced map

$$\lambda : \ G(\mathbb{A}_F)/G(F)Stab_{\mathbb{A}}(\mathbf{X}) \longrightarrow \prod_1^n \mathbb{I}_E \Big/ \left(\prod_1^n E^\times\right) \cdot \Xi$$

is injective.

The main result of this section is to show the existence of admissible subgroups.

Theorem 1.10. *If $\mathbf{X}$ is a separated scheme over o_F of finite type such that the generic fiber X_F is a trivial torsor of a multiplicative group G satisfying (1.2), then $\mathbf{X}$-admissible subgroups of $\prod_1^n \mathbb{I}_E$ always exist.*

Proof. By the conditions (1) and (2) for the choice of S_0, one has

$$\lambda[Stab(\mathbf{X}(o_{F_p}))] = \lambda[\mathbf{G}(o_{F_p})] = ker\left[\nu : \left(\prod_1^n o_{E_p}^\times\right) \longrightarrow \mathbf{T}(o_{F_p})\right]$$

for all $p \notin S_0$. For each $p \in S_0$, one can fix a positive integer r_p such that r_p kills all torsion points of $T(F_p)$. Let

$$r = \prod_{p\in S_0} r_p.$$

Let S_1 be a finite subset of Ω_F outside ∞_F such that Lemma 1.5 holds for $k = l \cdot r$. Define

$$\Xi = \left[\prod_{p\in S_1} \lambda(Stab(\mathbf{X}(o_{F_p}))) \cdot \mu([c \cdot 2t \cdot h(T, S_0) \cdot r]T(F_p))\right] \times \left(\prod_{p\notin S_1} \prod_1^n o_{E_p}^\times\right).$$

By Lemma 1.6, one has that Ξ is an open subgroup of $\prod_1^n \mathbb{I}_E$ and

$$\lambda[Stab_{\mathbb{A}}(\mathbf{X})] \subseteq \Xi.$$

Consider $\sigma \in G(\mathbb{A}_F)$ such that $\lambda(\sigma) = a \cdot i$ with

$$a \in \prod_1^n E^{\times} \quad \text{and} \quad i \in \Xi.$$

Then

$$v(a) \in \left[\prod_{p \in S_1} [c \cdot 2t \cdot h(T, S_0) \cdot l \cdot r] T(F_p) \times \prod_{p \notin S_1} \mathbf{T}(o_{F_p}) \right].$$

By Lemma 1.5, there is $u \in T(F)$ such that $v(a) = [l \cdot r](u)$. Since

$$v(a \cdot (\mu([r](u)))^{-1}) = v(a) - [l \cdot r](u) = 0,$$

one obtains that

$$a \cdot (\mu([r](u)))^{-1} \in \lambda(G(F)).$$

Let $i = (i_p)_{p \in \Omega_F}$. One can write

$$i_p = s_p \cdot \mu([c \cdot 2t \cdot h(T, S_0) \cdot r](n_p))$$

with

$$s_p \in \lambda(Stab(\mathbf{X}(o_{F_p}))) \quad \text{and} \quad n_p \in T(F_p)$$

for $p \in S_1$. This implies that

$$[l \cdot r](u + [c \cdot 2t \cdot h(T, S_0)](n_p)) = 0$$

for $p \in S_1$. Therefore $u + [c \cdot 2t \cdot h(T, S_0)](n_p)$ is a torsion point of $T(F_p)$ for each $p \in S_1$.

If $p \in S_0$, then r kills $u + [c \cdot 2t \cdot h(T, S_0)](n_p)$. One concludes that

$$\mu([r](u)) \cdot \mu([c \cdot 2t \cdot h(T, S_0) \cdot r](n_p)) = 1.$$

If $p \in S_1 \setminus S_0$, one has

$$[r]u + [c \cdot 2t \cdot h(T, S_0) \cdot r](n_p) \in \mathbf{T}(o_{F_p})$$

by the condition (4) for the choice of S_0. By the conditions (1) and (3) for the choice of S_0, one obtains that

$$\mu([r]u) \cdot \mu([c \cdot 2t \cdot h(T, S_0) \cdot r](n_p)) \in \prod_1^n o_{E_p}^{\times}.$$

Therefore one concludes that

$$\mu([r]u) \cdot i \in \lambda[Stab_{\mathbb{A}}(\mathbf{X})] \quad \text{and} \quad \sigma \in G(F) Stab_{\mathbb{A}}(\mathbf{X}).$$

The proof is complete. □

Since X_F is a trivial G-torsor over F, one can fix a rational point $Q \in X_F(F)$ which induces an isomorphism $X_F \cong G$ as F-varieties. Since $\mathbf{X}$ is separated over o_F, the natural morphism

$$\prod_{p \in \Omega_F} \mathbf{X}(o_{F_p}) \longrightarrow X_F(\mathbb{A}_F)$$

is injective. Define

$$f:\ \prod_{p\in\Omega_F}\mathbf{X}(o_{F_p})\to X_F(\mathbb{A}_F)\cong G(\mathbb{A}_F)\xrightarrow{\lambda}\prod_1^n\mathbb{I}_E.$$

Corollary 1.11. *Let Ξ be an $\mathbf{X}$-admissible subgroup of $\prod_1^n\mathbb{I}_E$. Then*

$$\mathbf{X}(o_F)\neq\emptyset\quad \textit{if and only if}\ f\left[\prod_{p\in\Omega_F}\mathbf{X}(o_{F_p})\right]\bigcap\left[\left(\prod_1^n E^\times\right)\cdot\Xi\right]\neq\emptyset.$$

Proof. Since $\mathbf{X}$ is separated over o_F, one has $\mathbf{X}(o_F)\subseteq\mathbf{X}(o_{F_p})$ for all $p\in\Omega_F$ and

$$\mathbf{X}(o_F)\subseteq\prod_{p\in\Omega_F}\mathbf{X}(o_{F_p})$$

by the diagonal map. If $\mathbf{X}(o_F)\neq\emptyset$, then

$$f\left[\prod_{p\in\Omega_F}\mathbf{X}(o_{F_p})\right]\bigcap\left[\left(\prod_1^n E^\times\right)\cdot\Xi\right]\supseteq f_E[\mathbf{X}(o_F)]\cap\left(\prod_1^n E^\times\right)\neq\emptyset$$

and the necessity follows.

Conversely, there is

$$y_A\in\prod_{p\in\Omega_F}\mathbf{X}(o_{F_p})\quad\text{such that } f(y_A)\in\left(\prod_1^n E^\times\right)\cdot\Xi.$$

By Definition 1.9, there are $\varrho\in G(F)$ and $\sigma_A\in Stab_A(\mathbf{X})$ such that $y_A=\varrho\sigma_A(Q)$. This implies that

$$\varrho(Q)=\sigma_A^{-1}(y_A)\in\prod_{p\in\Omega_F}\mathbf{X}(o_{F_p}).$$

Therefore $\varrho(Q)\in\mathbf{X}(o_F)\neq\emptyset$ and the proof is complete. □

If Ξ is an $\mathbf{X}$-admissible subgroup of $\prod_1^n\mathbb{I}_E$, there is an open subgroup Ξ_i of $\mathbb{I}_E$ for each $1\leq i\leq n$ such that

$$\prod_{i=1}^n\Xi_i\subseteq\Xi.$$

By class field theory, there is a finite abelian extension K_{Ξ_i}/E such that the Artin map

$$\psi_{K_{\Xi_i}/E}:\ \mathbb{I}_E/E^\times\Xi_i\cong Gal(K_{\Xi_i}/E)\tag{1.12}$$

gives the isomorphism for $1\leq i\leq n$. Projecting the image of f to the i-th component, one can define

$$f_i:\ \prod_{p\in\Omega_F}\mathbf{X}(o_{F_p})\longrightarrow\prod_1^n\mathbb{I}_E\longrightarrow\mathbb{I}_E$$

for $1\leq i\leq n$.

Corollary 1.13. *With the notation as above,* $\mathbf{X}(o_F) \neq \emptyset$ *if and only if there is*

$$x_A \in \prod_{p\in\Omega_F} \mathbf{X}(o_{F_p}) \quad \text{such that } \psi_{K_{\Xi_i}/E}(f_i(x_A)) = 1$$

in $Gal(K_{\Xi_i}/E)$ *for all* $1 \leq i \leq n$.

Proof. Since $\psi_{K_{\Xi_i}/E}(f_i(x_A)) = 1$ for $1 \leq i \leq n$, one has

$$f(x_A) \in \left[\prod_{i=1}^{n}(E^\times \cdot \Xi_i)\right] \subseteq \left(\prod_{1}^{n} E^\times\right)\cdot \Xi.$$

The result then follows by arguing as in the proof of Corollary 1.11. □

2. Brauer–Manin obstruction

In this section, we interpret **X**-admissible subgroups in terms of Brauer–Manin obstruction. This translation has been done for one dimensional tori in [16]. For the general case, the argument is similar. We keep the same notation as the previous section.

Since X_F is a trivial torsor of G over F, the fixed rational F-point Q gives a $Gal(\bar{F}/F)$-module homomorphism $\hat{G} \longrightarrow \bar{F}[X]^\times$, where $\bar{F}[X]^\times$ are the global sections of $\bar{X}$. Then one obtains a homomorphism

$$H^2(E, M) \cong H^2(F, \mathbb{Z}[Gal(E/F)] \otimes_{\mathbb{Z}} M) \to H^2(F, \hat{G}) \to H^2(F, \bar{F}[X]^\times) \tag{2.1}$$

by Shapiro's lemma. Applying the Hochschild–Serre spectral sequence (see Theorem 2.20 of Chapter III in [6]) in étale cohomology

$$H^p(F, H^q(\bar{X}, \mathbb{G}_m)) \Rightarrow H^{p+q}(X_F, \mathbb{G}_m),$$

one has

$$\phi_Q : H^2(E, M) \longrightarrow H^2(F, \bar{F}[X]^\times) \longrightarrow Br_1(X_F).$$

Fix a basis $\{e_1, \ldots, e_n\}$ of M such that the projection

$$p_i : \prod_{1}^{n} R_{E/F}(\mathbb{G}_m) \longrightarrow R_{E/F}(\mathbb{G}_m)$$

to the i-th component is given by $\mathbb{Z}e_i \subseteq M$ for each $1 \leq i \leq n$. Then the evaluation of the following morphism by using the fixed rational point Q

$$X_F \xrightarrow{Q} G \xrightarrow{\lambda} \prod_{1}^{n} R_{E/F}(\mathbb{G}_m) \xrightarrow{p_i} R_{E/F}(\mathbb{G}_m) \tag{2.2}$$

at $\prod_{p\in\Omega_F} \mathbf{X}(o_{F_p})$ is f_i in Section 1.

Let Ξ be an **X**-admissible subgroup of $\prod_1^n \mathbb{I}_E$ and Ξ_i be an open subgroup of $\mathbb{I}_E$ such that

$$\prod_{i=1}^{n} \Xi_i \subseteq \Xi$$

for each $1 \leq i \leq n$. Let K_{Ξ_i}/E be a finite abelian extension satisfying (1.12) for $1 \leq i \leq n$. Since

$$H^2(Gal(K_{\Xi_i}/E), \mathbb{Z}e_i) \cong H^1(Gal(K_{\Xi_i}/E), \mathbb{Q}/\mathbb{Z}) = Hom(Gal(K_{\Xi_i}/E), \mathbb{Q}/\mathbb{Z})$$

is finite, one has that the image of the inflation map composed with ϕ_Q

$$H^2(Gal(K_{\Xi_i}/E), \mathbb{Z}e_i) \to H^2(E, \mathbb{Z}e_i) \subseteq H^2(E, M) \xrightarrow{\phi_Q} Br_1(X_F)$$

is finite and denoted by $b(\Xi_i)$.

Definition 2.3. For any $\mathbf{X}$-admissible subgroup Ξ of $\prod_1^n \mathbb{I}_E$ and an open subgroup Ξ_i of $\mathbb{I}_E$ with

$$\prod_{i=1}^{n} \Xi_i \subseteq \Xi$$

for each $1 \leq i \leq n$, one defines the finite group $b(\Xi)$ of $Br_1(X_F)$ to be generated by $b(\Xi_i)$ for $1 \leq i \leq n$.

One can reformulate Corollary 1.11 in terms of Brauer–Manin obstruction.

Theorem 2.4. *Let Ξ be an $\mathbf{X}$-admissible subgroup of $\prod_1^n \mathbb{I}_E$. Then*

$$\mathbf{X}(o_F) \neq \emptyset \quad \textit{if and only if} \quad \left[\prod_{p\in\Omega_F} \mathbf{X}(o_{F_p})\right]^{b(\Xi)} \neq \emptyset.$$

Proof. The necessity follows from class field theory and one only needs to show the sufficiency. Applying Galois cohomology to the short exact sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow 0$$

with the trivial action, one obtains

$$\delta_i\, Hom(Gal(K_{\Xi_i}/E), \mathbb{Q}/\mathbb{Z}) \cong H^2(Gal(K_{\Xi_i}/E), \mathbb{Z}) \tag{2.5}$$

with $1 \leq i \leq n$. For any $\chi \in Hom(Gal(K_{\Xi_i}/E), \mathbb{Q}/\mathbb{Z})$, the cup product gives

$$\xi_i = e_i \cup \delta_i(\chi) \in H^2(Gal(K_{\Xi_i}/E), \mathbb{Z}e_i) \quad \text{and} \quad \beta_i = \phi_Q(\xi_i) \in b(\Xi_i)$$

with $1 \leq i \leq n$.

Let

$$(x_p)_{p\in\Omega_F} \in \left[\prod_{p\in\Omega_F} \mathbf{X}(o_{F_p})\right]^{b(\Xi)}$$

and evaluate β_i at $(x_p)_{p\in\Omega_F}$. Then

$$inv_p(\beta_i(x_p)) = \sum_{P|p} inv_P(f_i(x_p) \cup \delta_i(\chi))$$

by (2.1), (2.2), (8.1.4) Proposition and (7.1.4) Corollary in [9], where P's are all primes in E above p. One has

$$\sum_P inv_P(f_i(x_p) \cup \delta_i(\chi)) = 0$$

where P runs over all primes of E. This implies

$$\chi(\psi_{K_{\Xi_i}/E}(f_i[(x_p)_{p\in\Omega_F}])) = 0$$

for all $\chi \in Hom(Gal(K_{\Xi_i}/E), \mathbb{Q}/\mathbb{Z})$ by (8.1.11) Proposition in [9] and (1.12). Therefore

$$\psi_{K_{\Xi_i}/E}(f_i[(x_p)_{p\in\Omega_F}]) = 0$$

for $1 \le i \le n$. The result follows from Corollary 1.13. □

3. Strong approximation

As an application, we prove the strong approximation theorem for groups of multiplicative type. Such strong approximation theorem has been established for algebraic tori by Harari in [5] and generalized to connected reductive groups by Demarche in [4]. We give a different proof for groups of multiplicative type where G is not assumed to be connected. We keep the same notation as that in the previous sections.

By (8.1.16) Corollary in [9], the evaluation gives the pair

$$G(\mathbb{A}_F) \times H^2(F, \hat{G}) \xrightarrow{inv} \mathbb{Q}/\mathbb{Z} \tag{3.1}$$

such that $G(F)$ is lying in the left kernel of the above pair. Let $G(F_\infty)^0$ be the set of elements in $G(F_\infty)$ which are lying in the left kernel of (3.1).

Theorem 3.2. *$G(F)\cdot G(F_\infty)^0$ is dense in the left kernel of (3.1).*

Proof. By the definition of $G(F_\infty)^0$, one only needs to show that $G(F)\cdot U$ contains the left kernel of (3.1) for any open subgroup U of $G(\mathbb{A}_F)$ which contains $G(F_\infty)$. Without loss of generality, one can assume that

$$U = G(F_\infty) \times \prod_{p<\infty_F} U_p$$

where U_p is an open subgroup of $G(F_p)$ for all $p < \infty_F$ such that there is a finite subset S_0 of Ω_F outside ∞_F satisfying conditions (1)–(4) in Section 1 and

$$U_p = \mathbf{G}(o_{F_p})$$

for $p \notin S_0$. For each $p \in S_0$, one can fix a positive integer r_p such that r_p kills all torsion points of $T(F_p)$. Let $r = \prod_{p\in S_0} r_p$ and S_1 be a finite subset of Ω_F outside ∞_F such that Lemma 1.5 holds for $k = l\cdot r$. Define

$$\Xi = \left[\prod_{p\in S_1} \lambda(U_p)\cdot\mu([c\cdot 2t\cdot h(T,S_0)\cdot r]T(F_p))\right] \times \left(\prod_{p\notin S_1}\prod_1^n o_{E_p}^\times\right).$$

By Lemma 1.6, one has that Ξ is an open subgroup of $\prod_1^n \mathbb{I}_E$ and $\lambda(U) \subseteq \Xi$. Moreover the induced map

$$\lambda:\ G(\mathbb{A}_F)/G(F)\cdot U \longrightarrow \prod_1^n \mathbb{I}_E \Big/ \left(\prod_1^n E^\times\right)\cdot \Xi \tag{3.3}$$

is injective by the exact same arguments in Theorem 1.10.

Let Ξ_i be an open subgroup of $\mathbb{I}_E$ for each $1 \le i \le n$ such that

$$\prod_{i=1}^{n} \Xi_i \subseteq \Xi$$

and K_{Ξ_i}/E be a finite abelian extension such that (1.12) holds for $1 \le i \le n$. Then

$$0 \to \prod_{i=1}^{n} Hom(Gal(K_{\Xi_i}/E), \mathbb{Q}/\mathbb{Z}) \to \prod_{1}^{n} Hom(Gal(\bar{F}/E), \mathbb{Q}/\mathbb{Z}) \cong H^2(E, M).$$

Let

$$g = (g_p) \in G(\mathbb{A}_F) \quad \text{and} \quad \lambda(g) = (\sigma_1, \ldots, \sigma_n)$$

with $\sigma_i \in \mathbb{I}_E$ for $1 \le i \le n$. By the functoriality of evaluation and (8.1.11) Proposition in [9], one obtains

$$\sum_{i=1}^{n} \chi_i(\psi_{K_{\Xi_i}/E}(\sigma_i)) = inv(g, \lambda^*(\chi_1, \ldots, \chi_n))$$

by (3.1) for all

$$(\chi_1, \ldots, \chi_n) \in \prod_{i=1}^{n} Hom(Gal(K_{\Xi_i}/E), \mathbb{Q}/\mathbb{Z}).$$

If g is in the left kernel of (3.1), then $\sigma_i \in E^\times \Xi_i$ for $1 \le i \le n$. Therefore

$$\lambda(g) \in \left(\prod_{1}^{n} E^\times\right) \cdot \Xi.$$

The injectivity of (3.3) implies that $g \in G(F) \cdot U$. The proof is complete. □

When G is a finite commutative group scheme over F, the above result follows from Poitou–Tate (see (8.6.13) in [9]).

Remark 3.4. When $F_p = \mathbb{R}$, one defines

$$N_G(F_p) = \{x + \bar{x} : x \in G(\mathbb{C}) \text{ and } \bar{x} \text{ is the conjugate point of } x\}.$$

When $F_p = \mathbb{C}$, one defines

$$N_G(F_p) = G(F_p).$$

Let

$$N_G(F_\infty) = \prod_{p \in \infty_F} N_G(F_p).$$

By the local duality over $\mathbb{R}$ by (7.2.17) Theorem in [9], one has that

$$N_G(F_\infty) \subseteq G(F_\infty)^0$$

with the finite index. One can expect that $G(F) \cdot N_G(F_\infty)$ is dense in the left kernel of (3.1) by more careful study on archimedean places.

4. Sum of two squares

The natural extension of Fermat–Gauss' theorem about the sum of two squares over $\mathbb{Z}$ to the ring of integers of quadratic field F was already studied by Niven for $F = \mathbb{Q}\left(\sqrt{-1}\right)$ in [10] and

by Nagell for $F = \mathbb{Q}\left(\sqrt{d}\right)$ where

$$d = \pm 2, \pm 3, \pm 5, \pm 7, \pm 11, \pm 13, \pm 19, \pm 43, \pm 67, \pm 163$$

in [7,8]. Both followed Gauss' original idea. Therefore the class number one is assumed and the results obtained there always satisfy the local–global principle. In this section, we will apply the method developed in the previous sections to study this question and give an example for which the local–global principle is no longer true.

It should be pointed out that our method only produces the idelic class groups of $\mathbb{Q}\left(\sqrt{d}, \sqrt{-1}\right)$ for solving the problem of sum of two squares. In order to get the explicit conditions for the sum of two squares, one needs further to construct the explicit abelian extensions of $\mathbb{Q}\left(\sqrt{d}, \sqrt{-1}\right)$ corresponding to the idelic class groups by class field theory. Such explicit construction is a wide open problem (Hilbert's 12-th problem) in general. Some special cases can be constructed by the ad hoc method. In his series papers [14,15], the first author gives explicit construction for infinitely many d and solves the sum of two squares over infinitely many quadratic fields.

Let l be a prime with $l \equiv -1 \mod 8$ and $F = \mathbb{Q}\left(\sqrt{-2l}\right)$. We will study the sum of two squares over o_F. Let

$$E = F\left(\sqrt{-1}\right) \quad \text{and} \quad \Theta = E\left(\sqrt[4]{l}\right).$$

Lemma 4.1. *The field Θ/E is unramified over all primes except the prime above l.*

Proof. Since 2 is totally ramified in $E/\mathbb{Q}$, there is a unique prime v of E over 2. One only needs to show that v is unramified in Θ/E.

Since $\sqrt{l} \notin E$, one obtains that $x^4 - l$ is irreducible over E by the Kummer theory. By Hensel's Lemma, there is $s \in (1 + 4\mathbb{Z}_2)$ such that $s^2 = -l$ over $\mathbb{Z}_2$. Since

$$E_v = \mathbb{Q}_2\left(\sqrt{-1}, \sqrt{2}\right) = \mathbb{Q}_2(\zeta_8)$$

where ζ_8 is the primitive 8-th root of unity, one has

$$x^4 - l = x^4 + s^2 = x^4 - s^2\left(\sqrt{-1}\right)^2 = \left(x^2 - s\sqrt{-1}\right)\left(x^2 + s\sqrt{-1}\right)$$

over E_v. By $\zeta_8^2 = \sqrt{-1}$ and $-1 = \left(\sqrt{-1}\right)^2$, one concludes that Θ/E is unramified over v. □

It is clear that l is ramified in $F/\mathbb{Q}$ and the unique prime of F over l is also denoted by l. Since $l \equiv 3 \mod 4$, one has that l is inert in E/F. The unique prime of E above l is denoted by l as well and

$$E_l = \mathbb{Q}_l\left(\sqrt{-1}, \sqrt{l}\right).$$

Lemma 4.2. *If $N_{E_l/F_l}(\xi) = 1$, then the Hilbert symbols over E_l*

$$\left(\xi, \sqrt{l}\right)_l = \left(\xi, -\sqrt{l}\right)_l = 1.$$

If $N_{E_l/F_l}(\xi) = -1$, then the Hilbert symbols over E_l

$$\left(\xi, \sqrt{l}\right)_l = \left(\xi, -\sqrt{l}\right)_l = -1.$$

Proof. If $N_{E_l/F_l}(\xi) = 1$, then ξ is a unit of E_l. Therefore

$$\left(\xi, \pm\sqrt{l}\right)_l = \left(\xi, \sqrt{-2l}\right)_l \cdot \left(\xi, \pm\sqrt{-2}\right)_l = \left(N_{E_l/F_l}(\xi), \sqrt{-2l}\right)_l = 1$$

by 63:11a in [11] and (1.5.3) Proposition and (7.1.4) Corollary in [9].

If $N_{E_l/F_l}(\xi) = -1$, then

$$\left(\xi, \pm\sqrt{l}\right)_l = \left(\xi, \sqrt{-2l}\right)_l \cdot \left(\xi, \pm\sqrt{-2}\right)_l = \left(N_{E_l/F_l}(\xi), \sqrt{-2l}\right)_l = (-1, 2l)_l = -1$$

by 63:11a in [11] and (1.5.3) Proposition and (7.1.4) Corollary in [9]. □

Let $B = o_F + o_F\sqrt{-1}$ be the order of E and K_B be the ring class field of E defined by B.

Proposition 4.3. *Let* $\mathbf{X}$ *be the scheme defined by* $x^2 + y^2 = n$ *for some non-zero integer* n *over* o_F. *Then* $\mathbf{X}(o_F) \neq \emptyset$ *if and only if there is*

$$\prod_{w\in\Omega_F} (x_w, y_w) \in \prod_{w\in\Omega_F} \mathbf{X}(o_{F_w})$$

such that

$$\psi_{K_B/E}\left(\tilde{f}_E\left[\prod_{w\in\Omega_F} (x_w, y_w)\right]\right) = 1 \quad \text{and} \quad \psi_{\Theta/E}\left(\tilde{f}_E\left[\prod_{w\in\Omega_F} (x_w, y_w)\right]\right) = 1$$

where K_B *and* Θ *are defined as above, both* $\psi_{K_B/E}$ *and* $\psi_{\Theta/E}$ *are the Artin maps and*

$$\tilde{f}_E[(x_w, y_w)] = \begin{cases} \left(x_w + y_w\sqrt{-1}, x_w - y_w\sqrt{-1}\right) & \text{if } w \text{ splits in } E/F \\ x_w + y_w\sqrt{-1} & \text{otherwise.} \end{cases}$$

Proof. Define $E_w = E \otimes_F F_w$ for any $w \in \Omega_F$. Let B_w be the completion of B inside E_w for $w < \infty_F$ and $B_w = E_w$ for $w \in \infty_F$. Let

$$SO_{\mathbb{A}}(B) = \{\sigma \in R^1_{E/F}(\mathbb{G}_m)(\mathbb{A}_F) : \sigma B = B\}.$$

By Lemmas 4.1 and 4.2, the natural group homomorphism

$$\lambda_E : R^1_{E/F}(\mathbb{G}_m)(\mathbb{A}_F)/R^1_{E/F}(\mathbb{G}_m)(F)SO_{\mathbb{A}}(B) \to [\mathbb{I}_E/(E^\times N_{\Theta/E}(\mathbb{I}_\Theta))] \times \left[\mathbb{I}_E \Big/ \left(E^\times \prod_{w\in\Omega_F} B_w^\times\right)\right]$$

is well-defined.

If $u \in ker\lambda_E$, there are

$$\alpha \in E^\times \quad \text{and} \quad i \in \prod_{w\in\Omega_F} B_w^\times$$

with $\lambda_E(u) = \alpha i$. Therefore

$$N_{E/F}(\alpha) = N_{E/F}(i)^{-1} \in F \cap \left(\prod_{w\in\Omega_F} o_{F_w}^\times\right) = \{\pm 1\}.$$

Suppose $N_{E/F}(\alpha) = N_{E/F}(i) = -1$. Write

$$i = (i_w)_w \in \prod_{w\in\Omega_F} L_w^{\times}.$$

Since Θ/E is unramified over all primes of E except l by Lemma 4.1, one has that $\psi_{\Theta/E}(i_w)$ is trivial for all primes $w \neq l$, where i_w is regarded as an idele whose w-component is i_w and 1 otherwise. Since

$$N_{E/F}(i_l) = N_{E_l/F_l}(i_l) = -1,$$

one gets

$$\psi_{\Theta/E}(\alpha i) = \psi_{\Theta/E}(i) = \psi_{\Theta/E}(i_l) = -1 \in \mu_4$$

by Lemma 4.2, where μ_4 is the set of 4-th roots of unity and $Gal(\Theta/E) \cong \mu_4$. This contradicts to $u \in ker\lambda_E$.

Therefore

$$N_{E/F}(\alpha) = N_{E/F}(i) = 1.$$

This implies that

$$\alpha \in R^1_{E/F}(\mathbb{G}_m)(F) \quad \text{and} \quad i \in SO_{\mathbb{A}}(B).$$

One concludes that λ_E is injective. The result follows from Corollary 1.13. □

Finally we will give an explicit example by Proposition 4.3 such that the local–global principle is not true.

Let

$$N_{F/\mathbb{Q}}(\delta) = 2^{s_1}7^{s_2}p_1^{e_1}\cdots p_g^{e_g} \quad \text{and} \quad D(\delta) = \{p_1, \ldots, p_g\}$$

where $\delta = a + b\sqrt{-14}$ $(a, b \in \mathbb{Z})$ is an integer of $F = \mathbb{Q}\left(\sqrt{-14}\right)$. Denote $a = 7^{s_3}a_1$ with $a_1 \in \mathbb{Z}, 7 \nmid a_1$ and

$$D_1 = \left\{p \in D(\delta) : \left(\frac{-1}{p}\right) = \left(\frac{14}{p}\right) = 1 \text{ and } \left(\frac{7}{p}\right) = -1\right\}$$

$$D_2 = \left\{p \in D(\delta) : \left(\frac{-1}{p}\right) = -\left(\frac{14}{p}\right) = 1 \text{ and } \left(\frac{7}{p}\right) = -1\right\}$$

$$D_3 = \left\{p \in D(\delta) : \left(\frac{-1}{p}\right) = \left(\frac{14}{p}\right) = 1 \text{ and } \left(\frac{7}{p}\right)_4 = -1\right\}.$$

It is clear that e_i is even for $p_i \in D_2$.

Example 4.4. Let $F = \mathbb{Q}\left(\sqrt{-14}\right)$ and $\delta \in o_F$ as above. Then δ can be written as a sum of two squares over o_F if and only if

(1) δ can be written as a sum of two squares over o_{F_w} at every place w of F.
(2) $D_1 \neq \emptyset$; or

$$\left(\frac{a_1}{7}\right) = (-1)^{s_1 + \frac{s_2}{2} + \sum\limits_{p_i\in D_2}\frac{e_i}{2} + \sum\limits_{p_i\in D_3} e_i}$$

for $D_1 = \emptyset$.

Proof. Let H be the Hilbert class field of F. By Proposition 5.31 in [3], one has that $[H : F] = 4$. Since 2 is totally ramified in $E/\mathbb{Q}$ (see Exercise 6.7 in [3]), one has $H \cap E = F$. It is clear that $K_B \supseteq H.E$. By PARI/GP computation, one gets that the class number of the order $B = o_F + o_F\sqrt{-1}$ is 4. One concludes that $K_B = H.E$.

For any $\prod_{w\in\Omega_F}(x_w, y_w) \in \prod_{w\in\Omega_F} \mathbf{X}(o_{F_w})$, we have

$$\begin{aligned}\psi_{K_B/E}\left(\tilde{f}_E\left[\prod_{w\in\Omega_F}(x_w, y_w)\right]\right) &= \psi_{H.E/E}\left(\tilde{f}_E\left[\prod_{w\in\Omega_F}(x_w, y_w)\right]\right)\\ &= \psi_{H/F}\left(\prod_{w\in\Omega_F} N_{E/F}(\tilde{f}_E[(x_w, y_w)])\right)\\ &= \psi_{H/F}(\delta) = 1\end{aligned}$$

by class field theory. Therefore one only needs to consider the condition in Proposition 4.3 for $\Theta = E\left(\sqrt[4]{7}\right)$ where

$$iGal(\Theta/E) \cong \mathbb{Z}/4\mathbb{Z} = \{\pm 1, \pm\mathbf{i}\}$$

and $\mathbf{i}$ is the primitive 4-th root of unity.

If v is the unique prime in E above 2, then $E_v = \mathbb{Q}_2\left(\sqrt{-1}, \sqrt{2}\right)$ and v is unramified and splits into two primes in Θ/E. Then

$$\psi_{\Theta/E}(\tilde{f}_E[(x_w, y_w)]) = (-1)^{s_1}$$

where $w = v|_F$.

If v (resp. w) is the unique place of E (resp. F) over 7, then $E_v = \mathbb{Q}_7\left(\sqrt{-1}, \sqrt{7}\right)$, $F_w = \mathbb{Q}_7\left(\sqrt{-7}\right)$ and E_v/F_w is unramified. Moreover, one has

$$\sqrt{-1} = \frac{1}{2}\left(1 + \sqrt{-1}\right)^2 \in (E_v^{\times})^2.$$

By the local solvable condition at place w, one has that s_2 is even and

$$ord_w(a) < ord_w\left(b\sqrt{-14}\right).$$

Since v splits into two primes in Θ/E, one has the following Hilbert symbol

$$\begin{aligned}\psi_{\Theta/E}(\tilde{f}_E[(x_w, y_w)]) &= \left(\frac{(x_w + y_w\sqrt{-1}), \sqrt{7}}{E_v}\right)\\ &= \left(\frac{(x_w + y_w\sqrt{-1}), \sqrt{-7}}{E_v}\right) = \left(\frac{\delta, \sqrt{-7}}{F_w}\right)\\ &= \left(\frac{a, \sqrt{-7}}{F_w}\right) = \left(\frac{a, 7}{\mathbb{Q}_7}\right) = \left(\frac{a_1}{7}\right)(-1)^{\frac{s_2}{2}}\end{aligned}$$

by (1.5.3) Proposition and (7.1.4) Corollary in [9].

If $(p, 14N_{F/\mathbb{Q}}(\delta)) = 1$, then all primes above p are unramified in Θ/E. One has

$$\psi_{\Theta/E}\left(\tilde{f}_E\left[\prod_{w|p}(x_w, y_w)\right]\right) = 1.$$

If $p \in D(\delta)$ with $\left(\frac{-1}{p}\right) = -1$, then

$$F_w = \mathbb{Q}_p\left(\sqrt{-1}\right) = \mathbb{Q}_p\left(\sqrt{-14}\right)$$

by the local solvability at w with $w|p$. Therefore $\sqrt{7} \in F_w$. The Hilbert symbol

$$\left(\frac{p, \sqrt{7}}{F_w}\right) = \left(\frac{p, (-1)^t 7}{\mathbb{Q}_p}\right) = 1$$

by (1.5.3) Proposition and (7.1.4) Corollary in [9], where

$$t = \begin{cases} 0 & \text{if } \left(\frac{7}{p}\right) = 1 \\ 1 & \text{otherwise.} \end{cases}$$

This implies that $\sqrt{7} \in (F_w^{\times})^2$ by 63:12 in [11]. One concludes that each prime above w splits completely in Θ/E. Hence

$$\psi_{\Theta/E}(\tilde{f}_E[(x_w, y_w)]) = 1.$$

We restrict ourselves to $p \in D(\delta)$ with $\left(\frac{-1}{p}\right) = 1$.

If one further has $\left(\frac{7}{p}\right) = 1$ and $x^4 \equiv 7 \mod p$ is solvable, then all primes above p split completely in Θ/E. Hence

$$\psi_{\Theta/E}(\tilde{f}_E[(x_w, y_w)]) = 1$$

for $w|p$.

If $\left(\frac{7}{p}\right) = 1$ and $x^4 \equiv 7 \mod p$ is not solvable, then

$$\psi_{\Theta/E}(\tilde{f}_E[(x_w, y_w)]) = \begin{cases} \left(\frac{\delta, \sqrt{7}}{F_w}\right) = (-1)^e & \text{if } p \in D_3 \setminus D_1 \\ 1 & \text{otherwise} \end{cases}$$

where $e = ord_w(\delta)$ with $w|p$.

If $p \in D_2$, then p is inert in $F/\mathbb{Q}$ and splits completely into v_1 and v_2 in E. Moreover, each v_i with $i = 1, 2$ splits into two primes in Θ. Therefore

$$\psi_{\Theta/E}(\tilde{f}_E[(x_w, y_w)]) = \left(\frac{\delta, \sqrt{7}}{F_w}\right) = (-1)^{\frac{e}{2}}$$

with $w|p$ and $e = ord_p(N_{F/\mathbb{Q}}(\delta))$.

If $p \in D_1$, then p splits completely in $F/\mathbb{Q}$. Fix $w|p$ in F. One has that w splits into v_1 and v_2 completely in E/F. Let $\mathbf{i}$ be the image of the Frobenius of v_1 inside $Gal(\Theta/E)$. Then the image of the Frobenius of v_2 inside $Gal(\Theta/E)$ is $-\mathbf{i}$ by considering the action of $Gal(E/F)$. This implies that

$$\psi_{\Theta/E}(\tilde{f}_E[(x_w, y_w)]) = \mathbf{i}^a \cdot (-\mathbf{i})^b = (-1)^b \cdot \mathbf{i}^e$$

where $a = ord_w\left(x_w \pm y_w\sqrt{-1}\right)$ and $b = ord_w\left(x_w \mp y_w\sqrt{-1}\right)$ and $e = a + b = ord_w(\delta)$.

Now we summarize the above computation as follows. For any $p_i \in D(\delta)$, one has

$$\psi_{\Theta/E}\left(\prod_{w|p_i} \tilde{f}_E[(x_w, y_w)]\right) = \begin{cases} (-1)^{b_i} \cdot \mathbf{i}^{e_i} & \text{if } p_i \in D_1 \\ (-1)^{\frac{1}{2}e_i} & \text{if } p_i \in D_2 \\ (-1)^{e_i} & \text{if } p_i \in D_3 \setminus D_1 \\ 1 & \text{otherwise} \end{cases}$$

with $\prod_{w|p_i}(x_w, y_w) \in \prod_{w|p_i} \mathbf{X}(o_{F_w})$ and

$$b_i = \sum_{w|p_i} ord_w\left(x_w \mp y_w\sqrt{-1}\right).$$

Since $\delta \in N_{E_v/F_w}(E_v^\times)$ for $v|w|2$, one has

$$1 = \left(\frac{-1, N_{F_w/\mathbb{Q}_2}(\delta)}{\mathbb{Q}_2}\right) = \left(\frac{-1, n'}{\mathbb{Q}_2}\right) = \left(\frac{7, n'}{\mathbb{Q}_2}\right)$$

where $n' = p_1^{e_1} \cdots p_g^{e_g}$. Since $E_v = F_w\left(\sqrt{7}\right)$ and $\delta \in N_{E_v/F_w}(E_v^\times)$ for $v|w|7$, one has

$$1 = \left(\frac{7, N_{F_w/\mathbb{Q}_2}(\delta)}{\mathbb{Q}_7}\right) = \left(\frac{7, n'}{\mathbb{Q}_7}\right).$$

By the Hilbert reciprocity law (see 71:18 in [11]), one obtains

$$\prod_{i=1}^{g}\left(\frac{7}{p_i}\right)^{e_i} = 1.$$

If $\left(\frac{-1}{p_i}\right) = -1$ or $\left(\frac{-14}{p_i}\right) = -1$ for $p_i \in D(\delta)$, then e_i is even by the local conditions. Therefore we have

$$\sum_{p_i \in D_1} e_i \equiv 0 \mod 2$$

and $\psi_{\Theta/F}$ takes value $\{\pm 1\}$ over $\prod_{w\in\Omega_F} \mathbf{X}(o_{F_w})$.

When $D_1 \neq \emptyset$, then $\psi_{\Theta/E}(\tilde{f}_E[(x_w, y_w)])$ takes two different values by proper choice of the local solution $(x_w, y_w) \in \mathbf{X}(o_{F_w})$ for $w|p_i$ with $p_i \in D_1$. Hence there exists

$$\prod_{w\in\Omega_F}(x_w, y_w) \in \prod_{w\in\Omega_F} \mathbf{X}(o_{F_w}) \quad \text{such that } \psi_{\Theta/E}\left(\prod_{w\in\Omega_F} \tilde{f}_E[(x_w, y_w)]\right) = 1.$$

When $D_1 = \emptyset$, one concludes that $\mathbf{X}(o_F) \neq \emptyset$ if and only if

$$\left(\frac{a_1}{7}\right) = (-1)^{s_1 + \frac{s_2}{2} + \sum_{p_i\in D_2} \frac{e_i}{2} + \sum_{p_i\in D_3} e_i}$$

by Proposition 4.3. □

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